

OptBot: Toward the Most Accurate Sports Simulation Ever Built

What Is Proven, What Is Built, and What Remains

OptBot Research

Status Report v1.0 — July 2026 — Confidential

Abstract

OptBot projects what an NHL player becomes in an environment he has never played in — before a single shift is observed there. This report separates, without spin, what is *proven*, what is *built*, and what is *still to come*. The headline: on 860 real player–team transitions spanning July 2019 through the 2025 off-season — scored against independently licensed MoneyPuck expected-goals actuals, with every input frozen at each move date — the *minimal* version of the system already beats the industry-standard projection by 7.6% (RMSE 5.74 vs. 6.21; bootstrap 95% CI on the difference $[-0.61, -0.33]$, excluding zero), and its 80% uncertainty intervals achieve **80.0%, 80.0%, and 79.8%** measured coverage across player-experience bins. No neural network has yet been trained: these results come from audited statistics alone, and they define the floor — not the ceiling — of the system. The neural phases (A: environment tower; B: causal trunk; C: style vectors) are designed, coded, and gated, awaiting their first GPU hour.

1 What Is Proven

1.1 The headline results

Projector	RMSE (xGF%, first 40 GP)	vs. Marcel	95% CI	n
Last-season carryover	7.71	+24% worse	—	860
Marcel (industry standard)	6.21	—	—	860
OptBot v0	5.74	−7.6%	$[-0.61, -0.33]$	860

Companion result, in-house xG actuals, 2021–2025 ledger: v0 5.44 vs. Marcel 6.02 (−9.7%, CI $[-0.77, -0.40]$, $n = 546$).

Three properties make these numbers unusually hard to attack:

1. **Frozen-clock discipline.** Every projection uses only information strictly before its move date; an executable audit re-runs sampled moves with all post-date data physically deleted and asserts bit-identical output.
2. **Independent actuals.** The extended backtest is scored on MoneyPuck’s expected-goals model (used with permission), not our own — the referee is not on our payroll. The two rulers agree at $r = 0.925$ per window.
3. **Era transport.** The margin survives seven distinct eras, including thin-history 2019–20, COVID seasons, and 64 summer-2025 signings scored against the just-completed 2025–26 season. Edges that are artifacts collapse out-of-era; this one traveled.

1.2 Calibration: the result behind the result

Every projection ships with an 80% interval fit by split-conformal inference on historical moves. Measured coverage: **0.800 / 0.800 / 0.798** across rookie, mid-career, and veteran bins. To our knowledge no commercial sports-projection vendor publishes measured calibration at all; the predecessor system this project replaced achieved 12% at the same nominal level. Accuracy wins arguments; *measured honesty* wins contracts.

1.3 Distance to the theoretical ceiling

Forty games of on-ice xGF% carries an irreducible sampling noise floor of roughly 4 RMSE points — no model, however perfect, can beat it. Against the *closable* gap (Marcel at 6.21 down to the ~ 4.0 floor), v0 alone — one scalar per player plus a destination-environment average — has already captured $\approx 21\%$. The remaining ~ 1.7 points are the neural phases’ hunting ground, and the reason “most accurate sports simulation ever built” is a trajectory with a measured starting point rather than a slogan.

1.4 The data substrate (what “certified” means)

The training surface (6.5M+ player-windows, ten seasons collected) passed five independent certifications: (i) machine-checked internal contracts; (ii) external reconciliation vs. MoneyPuck (95.9% exact 5v5 goal agreement over 8,265 games); (iii) NHL-API spot audit (94–98% presence/event/bounds agreement); (iv) the original builder’s own zero-issue full-season token audits, rediscovered and adopted as regression gates; (v) a full-season event microscope against official shift charts: 99.7% of 986,920 player-window second-counts exact to $\leq 1s$, **100.00%** of 754,104 personal-event attributions correct, and **zero** goals mis-credited across 59,807 on-off-on substitution cases. Separately, all 617 lagged prior columns passed an empirical prediction-symmetry test (worst leak-gap 0.005 vs. 0.902 for a deliberate-leak control): the priors are *provably* lagged, not assumed lagged. Twenty-two named causal threats are tracked in a public register; each is closed, measured, priced into the intervals, or refused at the API. Fourteen silent data corruptions — including an outcomes column that was identically zero across 761,823 rows — were found and repaired along the way; several fixes *improved* the headline (cleaning the test set raised the measured edge from -8.6% to -9.7%).

2 What Is Built But Not Yet Trained

v0 is deliberately minimal: destination-environment expectation plus a deconfounded, empirical-Bayes-shrunk talent scalar. It cannot see linemates, lever settings, or playing style. The machinery that can — the actual Causal Intervention Network — is coded, unit-tested, and gated, but has consumed **zero GPU-hours**:

Phase	Role	Status	Ship gate
A — Environment tower	encode <i>who is on the ice</i>	coded, untrained	5-probe leakage scorecard
B — CIN trunk	levers, talent channel, outcomes	coded, untrained	beat v0 in the 860-move harness
C — Style vectors	fit & chemistry (13,708 shot-personality cards precomputed)	priors built, vectors unstated	probe-gated; v2

Supporting assets already banked for these phases: a per-shot attachment of MoneyPuck’s full shot model to our windows (99.86% match; boundary-second assignment ruled by event order

— a 7% -of-shots bug class caught before it shipped); rush / rebound / chaos / pressure target decompositions; entry-mode context (on-the-fly entrants measured at +12% xGF/60 — a selection effect the tower conditions on); and a feature-legality schema whose banned list cites the code trace that convicted each leaky column.

3 What Remains

1. **First GPU run** (Phase A tower + Phase B trunk, era-split): the per-phase feature ablations happen here, empirically, judged by the harness and the probes — legality was decided by trace; utility is decided by experiment.
2. **The shootout:** v1 must beat v0's 5.74 with a CI clear of zero, beat its own gradient-boosted twin, and pass all probes — or v0 ships. The company does not depend on the neural result; it is upside with a floor.
3. **Phase C v2:** style vectors and pairwise fit, unlocked only after A+B — the chemistry claims will be gated by an identifiability census, not enthusiasm.
4. **September:** projections for every 2026-27 mover, hash-sealed before puck drop. **October:** the sport watches them score in public. That — accuracy, performed live, with measured error bars — is the world-shocking act; the numbers above are what make it safe to attempt.

One-Sentence Summary

Before training a single neural network, OptBot already beats the sport's standard projection method by 7.6% on 860 frozen-clock trades scored by an independent referee, with uncertainty intervals accurate to two-tenths of a percent — and every component that could make it dramatically better is built, audited, and waiting for its first GPU hour.